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Multi-objective optimization under uncertainty of geothermal reservoirs using experimental design-based proxy models

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ABSTRACT

Geothermal energy has a high potential to contribute to a more sustainable energy system if the associated economic risks can be overcome in the design process. The development planning of deep geothermal reservoirs (over 1000 m depth) relies on computer models to forecast and then optimize system design. Optimization is easy where all the objective's (e.g. NPV) optimization parameters and, most importantly, the geology are considered as known, but this is almost always not the case. Where the complex engineering design (e.g. well placement) meets significant geological uncertainty every development option should be tested using an expensive simulation against the range of geological possibilities. The impracticality of simulating so many models results in a limited exploration of geological uncertainties and development options. Consequently, the risk of improper system design cannot be properly assessed. This paper presents an approach to understand the trade-offs in maximizing heat extraction while minimizing energy usage in re-injection for a new geothermal reservoir development while considering the uncertainty from 18 different geological models. Our approach is computationally feasible because we apply multi-objective particle swarm optimization (MOPSO), to an ensemble of response surface models, built using Gaussian process regression (GPR), for each and every geological scenario. MOPSO explores the trade-off surface for the competing objectives using the mean reservoir responses (covering the geological uncertainty). Our results highlight the impact of geological uncertainty on the optimal well placement and show the need to consider geological uncertainties adequately in optimization. The work demonstrates the shortcomings of using only one geological model of a geothermal reservoir and/or a single objective in optimization. We additionally demonstrate the practicalities of using response surface models in this way for geothermal systems. We anticipate that our work raises awareness for the scope of optimization of geothermal reservoir design under geological uncertainty.

1. Introduction

Faced with the imminent violation of the Paris Agreement's goals and the corresponding ecological consequences, geothermal energy is witnessing a renaissance of interest in the energy market (Pinzuti, 2017) as a green energy source capable of delivering constant (base load) energy output. Worldwide, high enthalpy resources are most suitable for power generation (Bertani, 2012). However, access to high enthalpy resources is restricted to comparably few places on earth like active volcanic regions. Most unexploited geothermal potential remains in low enthalpy reservoirs (i.e. reservoirs with temperatures below 150 °C, Haenel et al., 1988). Despite its potential for widespread use and low operational costs, low enthalpy projects struggle because of high investment costs yielding small profit margins at significant risk. The main cost driver, drilling costs, increases with depth (Lukawski et al., 2014, 2016) yet projects must achieve depths sufficient to generate enough heat. Risks emerge from a lack of geological subsurface data creating a broad set of geological uncertainties in forecasts of energy output. As a result, today's geothermal low enthalpy power plants operate with few boreholes (typically doublets), where success depends on a careful choice of their location and operation.

Areas with a sufficient amount of subsurface data from past hydrocarbon exploration can allow for probability of success studies to assess the geological risk for geothermal projects (Schumacher et al., 2020). Even so, decision making in geothermal projects mostly proceeds from computer models to predict reservoir performance over

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time and model prolonged power plant use. Numerical models solve both the fluid flow problem and the thermodynamics of heat transfer from the rock to the fluid, most often based on a finite difference, finite volume or finite element scheme. The coupled thermo-hydraulic processes make the simulation numerically more complex. Complexity adds to model run times enforcing the modeler to trade-off between model fidelity and the number of model realizations run to cover uncertainty. A single model reflects the behavior of only one possible geological outcome among many. Therefore, uncertainties are better represented by many rolls of the dice, i.e. by considering a wider range of geologically plausible scenarios (i.e. geological ideas/interpretations that can be described by a model).

If we optimize any reservoir development plan through reservoir simulation (well placement and control, use of pumps, etc.), and significant uncertainty exists, the price of optimizing under this uncertainty is an exponential increase in the computational cost. Covering all model possibilities requires a run for each geological scenario against every development plan. Where significant heterogeneity exists in properties such as porosity and permeability there also exists significant uncertainty in its spatial distribution; different spatial distributions of permeabilities leading to different flow paths, in turn leading to different engineering challenges to develop the reservoir. Geostatistical modeling (Demyanov and Arnold, 2018) has long been used in oil and gas reservoirs to model porosity/permeability heterogeneity and capture the impact of facies/fracture distributions on flow, breakthrough times and recoveries. The optimal development plan for one geological model can differ to an optimal plan for an alternative, but equally probable geological scenario. Therefore, optimization must take uncertainty into account.

Optimization under uncertainty successfully improved the performance of petroleum reservoirs where heterogeneity heavily influences flow of the multi-phase hydrocarbon systems present (Ali Ahmadi et al., 2012; Aanonsen et al., 1995; Chen et al., 2012). Uncertainty quantification approaches have been applied to geothermal reservoirs, most recently using a limited number of simulation forecasts to build statistical models or response surfaces, which are used to predict reservoir performance (Ansari and Hughes, 2016; Quinao and Zarrouk, 2018). Optimization under uncertainty has been shown to provide useful outcomes in petroleum reservoirs using similar response surface methodologies (Agada et al., 2017) or dimensional reduction techniques (Scheidt and Caers, 2008; Li et al., 2014; Arnold et al., 2016) to reduce compute time. Robust optimization has also been used in non-petroleum problems such as carbon capture and storage (CCS) projects (Petvipusit et al., 2014). For example, Lu et al. (2018) describe two approaches capturing geological uncertainty in the optimization process for geothermal reservoirs to optimize each geological scenario separately or to take the mean response across all geological scenarios. In their work they limited the problem to 8 simple geological models with spatially homogeneous properties to make the problem more computationally manageable for a wider range of development plans (multiple well pairs distributed across the reservoir), from which they estimate reservoir production using the superposition principle as a proxy to a full reservoir simulation.

A further dilemma in optimization arises where trade-offs exist between competing objectives. A plan adjusted to optimize one outcome (e.g. increased heat production rate), may worsen another (e.g. cold-water breakthrough) such that we cannot optimize all objectives together and the task instead becomes an exploration of the trade-off surface between them. Multi-objective optimization (MOO) comprises a family of algorithms that explore the trade-off surface known as the Pareto front rather than seek a single "best" (minimum/maximum) outcome; again we can draw on a range of examples from petroleum reservoirs that demonstrate their benefit (Hutahaean et al., 2016, 2018; Schulze-Riegert et al., 2007; Fonseca et al., 2015; Christie et al., 2013; Mohamed et al., 2011).

In this paper, we present one way to optimize a geothermal reservoir under geological uncertainty using a combination of MOO algorithms to explore engineering trade-offs, and a response surface approach trained on numerical simulations from a wide range of geological scenarios to forecast reservoir behavior and drive the optimization algorithm. Response surfaces aim to predict the model response without running the numerical simulation and thus benefit from much faster run times, but suffer from an unknown forecast error. Based on the sample of model forecasts we can afford computationally, even a good fitting response surface may not provide equally good predictions and the prediction errors will vary in an unknown and unpredictable way. In particular, the model complexity and the number of simulation responses used to train and tune the response surface has a significant impact on the functional fit and its predictability (Zubarev, 2009).

Our work generates a number of response surface models, based on reservoir simulations chosen by an experimental design technique (Box and Draper, 1986) to capture uncertainty, for a range of different geological concepts. This is a common approach to uncertainty analysis in petroleum reservoirs for a single geological scenario (e.g. White et al., 2000). As each geological concept is typically a discrete entity, we represented each as a scenario with its own response surface model, which considers uncertainty of fault transmissibility around that concept and a range of design parameters for the development scheme (in this case the placement and operation of a doublet). The work results in an ensemble of response surface models, each representing a geological end member that is possible according to geophysical exploration and well data. The ensemble of response surface models is then used by the MOO algorithm to explore the engineering trade-offs for a geothermal doublet where we wish to maximize energy extraction while minimizing re-injection energy (pumping). This complete approach of considering uncertainties along with engineering trade-offs in the optimization of the reservoir originates from recent work in hydrocarbon reservoirs (Yang et al., 2013), but to our knowledge it is new to geothermal applications.

2. Methodology

Simulation models predict the response of a geothermal reservoir when in operation, allowing researchers to both characterize uncertainty by varying the unknown geological parameters and to see their impact or to optimize the system by varying well and operations parameters to identify the best reservoir development design. Both uncertainty quantification and optimization necessitate a search of the model parameter space, the volume of which depends on the complexity of the model (i.e. the number of parameters). For optimization we aim to discover the best overall engineering solution for a given geology. For uncertainty quantification we focus on the discovery of many different geological ideas (combinations of parameters) that fit data and provide a wide range of forecasts for future production performance. The critical issue for both tasks is the computational cost of searching the parameter space as each sample involves running a complex and therefore slow computer model. The dimensionality (i.e. the number of uncertain parameters) of the problem and the complexity of the model response (i.e. models with similar parameter values but with vastly different responses) dictate the number of model runs required. However, computational run time is typically the main limitation of a study.

2.1. Geological uncertainty

Uncertainty exists because of a lack of enough precise data for a whole reservoir. Geothermal developments often run on much tighter budgets and have much smaller financial yields compared to their petroleum equivalents. Thus, geothermal projects often cannot afford to spend as much on data acquisition. Indirect measurements like seismic surveys lack the resolution required to see all scales of heterogeneity, but constitute the bulk of available spatial data as it is the only economically feasible way to image the subsurface volume (Takahashi, 2000). Furthermore, their interpretation is usually not unique (Mavko and Mukerji, 1998), compounding the degree of uncertainty. Outcrop analogues provide insights and inform model building, but the data differs from the reservoir conditions (Alexander, 1993). Consequently, interpretations include bias, which creates the need for a range of geological models.

The key geological uncertainties depend on the geological setting but include the localized geothermal gradient and rock heat capacity, faulting and barriers to flow, aquifer strength and recharge, and heterogeneity in the permeability field (from the pore network or fractures). Geological uncertainty arises from heterogeneity at various geological scales, which have significant effects on subsurface flow simulations (Eaton, 2006). Fuchs and Balling (2016a,b) showed the impact of geological heterogeneity on the spatial distribution of temperature within the North German basin, differences that are often attributed to convection or other heat processes rather than spatial geological variations. At the reservoir scale, Crooijmans et al. (2016) argue the importance of facies heterogeneity on low-enthalpy sedimentary reservoirs, showing the impact of Net-to-Gross (NTG) on production temperature. Willems et al. (2017a) show how connectivity and channel orientation impact upon doublet performance when more complex and realistic geological models of fluvial systems are used.

Uncertainty quantification techniques are widely used within the oil and gas industry where the impact of spatial uncertainties are estimated, particularly Bayesian approaches which use available production or seismic data to infer statistical estimates of uncertainty or update initial estimates. Computationally, this requires a trade-off between model complexity and computational power applied to the problem. The former is typically solved via some kind of reduced order modeling to capture the predictability in a simpler, quicker model. Examples of this in geothermal reservoirs include Mudunuru et al. (2017) who develop reduced order models (response surface models) based on polynomial regressions to 3D simulation data, and Khait and Voskov (2018) who used applied novel linearization techniques to improve the efficiency of solving the thermal model on a complex geological scenario.

2.2. Optimization

To speed up the process of optimization, a wide variety of algorithms exist to search the large volumes of parameter space. Stochastic optimization methods are a commonly used set of approaches whose main advantage is to prevent entrapment. Entrapment occurs when the algorithm gets stuck in one part of parameter space in a local minimum/maximum, one of many “good” regions of parameter space. The stochastic element of these algorithms allows it to jump out of local optima and increase the diversity of the ensemble. Stochastic techniques are extremely diverse, including commonly used approaches like particle swarm optimization (PSO) (Mohamed et al., 2010), differential evolution (Hajizadeh et al., 2010), genetic algorithms (Christie et al., 2006) and Bayesian optimization algorithms (Abdollahzadeh et al., 2012). PSO, for instance, has been used extensively in the optimization of hydrocarbon reservoir developments in both single-objective (Mohamed et al., 2010) and multi-objective versions (Christie et al., 2013). Many stochastic optimizers include a multi-objective version for problems where multiple competing objectives exist and their trade-off needs exploration. Multi-objective algorithms operate to discover a subset of sampled models that are better in at least one objective than all others; a subset called the Pareto front (Teich, 2001). Thus, the concept of *optimal* takes on a different meaning for multi-objective methods where the aim becomes (Zitzler, 1999):

1. minimize the distance between solutions on the Pareto front,
2. maximize the spread along the Pareto front,
3. maximize the number of elements on the Pareto front.

This work uses the multi-objective particle swarm optimization (MOPSO) algorithm (Mohamed et al., 2011), which includes additional steps from standard PSO to fulfil the above aims. In contrast to standard PSO, where the global best value is the best single objective value seen by any particle at the latest iteration, MOPSO updates the global best value throughout by one of the existing Pareto front model with a high (within the top 10% of all Pareto front models) separation from other Pareto models. This effectively fills the gaps between points fulfilling aim (1) and pushes the Pareto front wider fulfilling aim (2).

First MOPSO calculates the distances between the models on the Pareto front, measured by the *crowding distance*, which are ranked from high to low. The model with the largest crowding distance becomes the *leader*, equivalent to the global best in standard PSO, which then drives sampling to fill in this part of the Pareto front. The models at the extremes of the Pareto front with the smallest and largest values of the objective functions are given infinite crowding distances so they will continually draw samples to widen the frontier.

When the number of objectives increases beyond two the performance of MOO algorithms drops, as shown in Hutahaean et al. (2017). Hutahaean et al. (2017) show a way to reduce the number of objectives through grouping objectives together to minimize the conflicts between objectives within each group. More recent algorithms like reference vector-guided evolutionary algorithms show improved performance in exploring the Pareto front cases where many objectives are required (Hutahaean et al., 2016).

Optimization under uncertainty (OUU) requires further computational effort as we must test each development option against each geological scenario to ensure the chosen development plan will work no matter what the geology. Uncertainty quantification creates large ensembles of models with long run times. Thus, OUU is only practical if we can either speed up forecasting or pick a subset of geological scenarios. A wide range of approaches exist for reducing the ensemble including dimensional reduction and clustering techniques (Scheidt and Caers, 2008; Arnold et al., 2013). Reducing model run time requires a simpler model, achieved through coarsening the grid (reduced model fidelity), reduced physics (Josset et al., 2015) or replacing the simulation by a response surface (Ansari and Hughes, 2016).

2.3. Work flow

In this work we apply MOPSO to optimize the placement and operation of a geothermal doublet in a low enthalpy permeable reservoir under geological uncertainty. This is achieved through a combination of numerous reservoir simulations for a range of geological scenarios of the same field. To capture the range of geological outcomes we build nested ensembles of discrete proxy models that we call response surfaces. They represent the different geological concepts, and predict responses for the key metric of outlet temperature/enthalpy and bottom-hole pressure (BHP) in both wells of a geothermal well doublet.

A full factorial experimental design approach (Siebertz et al., 2010) is used to select the simulation models for proxy training. The simulation models forecast the outlet temperature and BHP over time to represent the net-energy output from a selected number of development plans of the system. Building a proxy model to predict time series data is particularly challenging, thus we build upon successful techniques taken from the petroleum industry described in Christie et al. (2006), Yu et al. (2016), Khaksar Manshad et al. (2016).

Firstly, a best fit linear regression model is found for each training simulation's time series response to allow for the time series data to be described parametrically. In total, five regression parameters are required to recreate each simulated time series: one for the initial/starting reservoir temperature, one to define the length of the plateau period of initial heat production and three parameters to describe the regression exponents to fit the decline of outlet temperature post-plateau, as illustrated in Fig. 1. An additional proxy is fitted to the difference in injector and producer bottom hole pressure (dBHP) to account for the energy required for

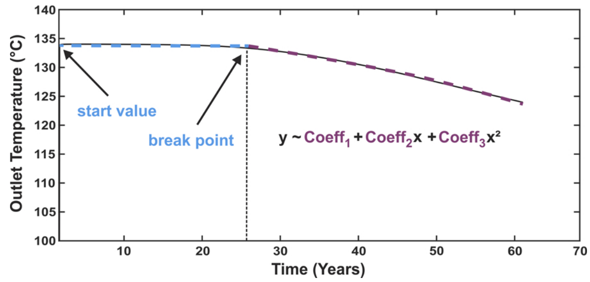


Fig. 1. Five parameters describing an exemplary time series of outlet temperature as a function of continuous geothermal production: *start value* and *break point* define the temperature and length of the initial plateau phase until cold water break through; coefficients $Coeff_1$, $Coeff_2$ and $Coeff_3$ define the declining temperature curve after cold water break through.

re-injection. Thus we build six proxy models per training simulation (one for each of the regression model parameters) to predict outlet temperature and pressure.

For each geological scenario Gaussian process regression (GPR) (Rasmussen, 2004; Samui and Kim, 2013) is applied on the respective proxy models to create the according response surface, a kernel-based approach to solving complex non-linear regression problems. A Gaussian processes is a generalization of a Gaussian distribution specified by a mean function $m(x)$ and a covariance function $k(x, x')$, which are parameterized in terms of a set of hyper-parameters, then tuned to train the GPR model. The precise kernel function used to describe the covariance is a modeler's choice. In this study we applied only a squared exponential kernel and as such the paper results are not aimed at demonstrating the ideal setup of GPR for proxy modeling. We focus instead on the impact of optimization under uncertainty, but we expect the kernel choice will impact on prediction performance.

Each response surface model can then predict the respective proxies' coefficients and allows for production forecasts of development plans that were not part of the training simulations. Finally, MOPSO can be run on the response surface models to explore the engineering trade-offs on a particular geological model or on the mean response of all response surfaces to consider geological uncertainty.

In summary, the generalized work flow (see Fig. 2) consists of the following steps:

1. Generate geological scenarios which cover the key geological uncertainties.
2. Use experimental design to select and test a range of development plans on each of the geological scenarios. Forecast the reservoir models.
3. Fit a proxy model using linear regression to each simulation of the experimental design and predict the time series.
4. Apply GPR on the subsets of respective proxies to create a response surfaces for each geological scenario.
5. Apply multi-objective optimization (MOPSO) to the response surfaces and explore the engineering trade-offs.
6. Evaluate each proposed development plan across all response surfaces (i.e. all geological scenarios) to estimate the mean response and the variance of the option.

3. Case Study

This work uses parts of the semi-synthetic Watt Field data set (Arnold et al., 2013) to represent the typical region occupied by a geothermal doublet. The Watt Field example was originally considered as a hierarchical benchmark case study for hydrocarbon reservoirs, designed specifically to consider interpretational uncertainty in the modeling work flow by providing a wide range of interpretations through combinations of key geological features. The following sections describe the nature of the Watt Field models, how we adapted the Watt Field to a geothermal problem and the operational design parameters of the doublet.

3.1. The Watt Field

The Watt Field (Arnold et al., 2013) has over 81 geological models that cover interpretational uncertainty integrated throughout the reservoir modeling workflow in a hierarchical way. The uncertainties include: different model grids, depth and shape of the reservoir's top surface, facies models, porosity and permeability models, and fault locations. Watt is a clastic, braided fluvial reservoir, forming a shallow anticline of 12.5 km by 2.5 km with a maximum relief of 190 m from the oil/water contact. This low relief makes the definition of the reservoir's confining surfaces a key uncertainty given its poor spatial control by data from 8 appraisal wells. The field is structurally dominated by a range of major E/W trending and associated N/S extensional faults whose precise number and location are unknown, thus resulting in 3 different interpretations of the fault network. For or our approach, we selected only 18 of the 81 geological scenarios to constrain the problem to a manageable size covering the following uncertainties (abbreviations are later used to distinguish models):

- **Top surface model (TS):** The form of any seismic horizon depends

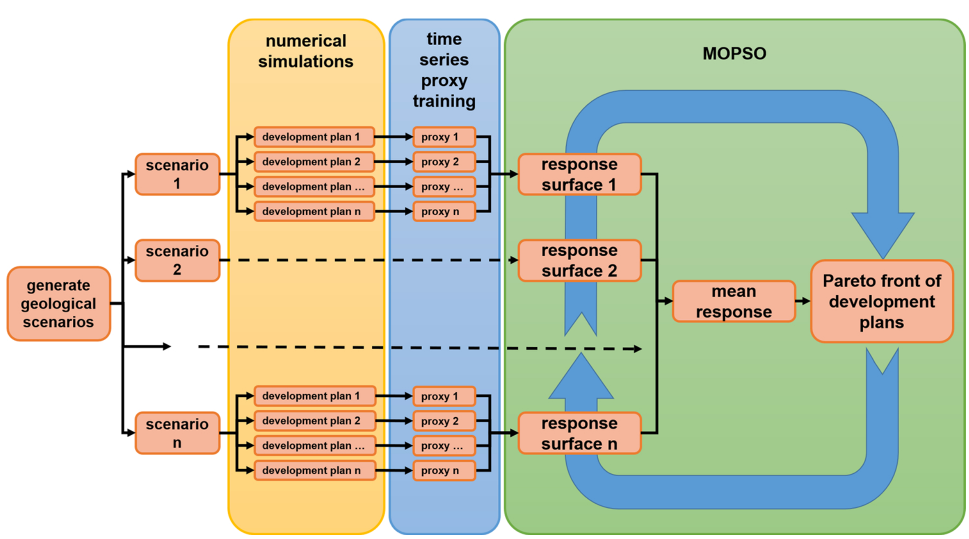


Fig. 2. Generalized work flow of proxy training, building of response surfaces and subsequent MOO.

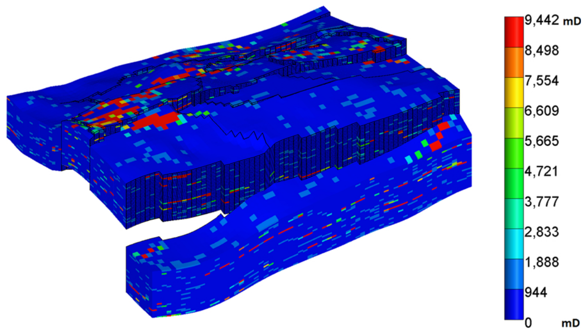


Fig. 3. Heterogeneous permeability distribution of a particular geothermal Watt Field model realization.

upon the choices in picking and smoothing the surfaces from seismic data and calibrating to wells. Three different top surface models are included in our case study that represent different interpretations of seismic data. The shape of the reservoir's top surface defines the volume of the reservoir and thus the available amount of heat in place.

- **Net-to-Gross model (CO):** A variety of net and non-net facies (e.g. shales, channel sands, overbank and crevasse sands and silts) exist in this reservoir, distinguished by the magnitude of permeability data and the proportion of shaley material (VShale). We applied three different NTG models, with Vshale cut-off levels of 0.6, 0.7 and 0.8 respectively to adjust the proportion of net sand. These cut-offs control the distributions of permeability in the reservoir (Fig. 3). As the flow paths in the reservoir depend on this distribution, for a given doublet it can have a strong influence on the time to cold water breakthrough and thus on the geothermal resource's lifetime.
- **Fault network model (FM):** The Watt Field data set provides three different fault network models. We used the two most dissimilar ones in our study. The first one includes mostly faults trending E-W, whereas the second one has additional faults trending N-S. As faults can represent barriers to fluid flow if sealed, they can influence flow paths and delay cold water breakthrough depending on the fault transmissibility.

3.2. Modeling scenario

A 6.6 × 4.5 km wide part of the Watt Field models was cropped from the original model, deepened to 4200 m depth for a shift to conditions of a typical deep geothermal low enthalpy reservoir and then refined to better resolve the thermal front as it moves through the reservoir. The initial pressure and temperature increased to match the new reservoir depth while porosity and permeability decreased to account for compaction. We assumed a normal geothermal gradient of 0.03 °C/m and a normal hydrostatic gradient of 9.8 kPa/m producing reservoir temperatures between 130 and 145 °C and pore pressures ranging from 39 to 44 MPa. The model does not consider an external aquifer and regards the model edge as a no-flow boundary.

One vertical injection well and one vertical production well, each with a diameter of 12.22 cm ($9\frac{5}{8}$ in.), intersect the reservoir across its entire thickness of 160 m. The model forecasts 60 years of operation using time steps of 31 days and maximum injection and production rates of 138.9 L/s (12,000 m^3/day). This provides enough time to reveal the impact of each uncertainty on production temperature/enthalpy rate and represents a likely operational lifetime. This is important, as the proxy models and response surfaces will only predict thermal depletion if they were trained on respective models. On shorter time spans many development plans can appear to perform equally well. To better distinguish the viability of different development plans, simulating 60 years operation makes sure most plans would see significant thermal depletion. More importantly, the scenario resembles the design and the conditions of a plausible geothermal low enthalpy reservoir (Franco and Villani, 2009).

Table 1
Key parameters of the Watt Field model.

Parameter	Value	Unit
Scale	6600 × 4500 × 200	m
Resolution	100 × 100 × 5	m
Number of cells in the model	115,830	–
Geothermal gradient	0.03	°C/m
Rock compressibility	1e−7	1/kPa
Rock heat capacity	1.82e6	J/(m^3 °C)
Rock thermal conductivity	2.64	W/(m °C)
Maximum injection rate	138.9	L/s
Maximum production rate	138.9	L/s
Maximum bottom hole pressure, injector	68,000	kPa
Minimum bottom hole pressure, producer	29,500	kPa
Mean porosity	0.17	
Mean permeability	465	mD

In order to avoid borehole collapse or induced reservoir fractures, the producer is limited by a minimum BHP of 29.5 MPa, or 70% of the undisturbed hydrostatic pressure at 4300 m depth. The injector is restricted by a maximum BHP of 68 MPa, which corresponds to approximately 70% of the lithostatic pressure at 4300 m depth assuming a lithostatic pressure gradient of 22.6 kPa/m. A minimum production temperature threshold was not imposed as it would strongly depend on the actual heat utilization cascade, which is beyond the scope of this study. Table 1 summarizes the model setup in more detail.

3.3. Variable model parameters

The model parameters fall into two groups: the geological uncertainties and engineering design parameters. Geological uncertainty is covered by the 18 combinations of the previously mentioned models (3 TS × 3 CO × 2 FM). Additionally, we included a multiplier for transmissibility across the faults as an additional geological uncertainty, which is set the same for all faults. However, transmissibility is not a discrete entity and therefore does not fundamentally change geological concepts described by combinations of the aforementioned models. Thus, additional discretizations are not required to consider transmissibilty. Instead, it can be imposed on a geological scenario as a variable parameter. Training simulations are run with faults either 1% or 90%, i.e. almost entirely closed or mostly open to flow.

Likewise, a variable skin factor could be included as an additional geological uncertainty for small scale effects at the wells like poor injectivity and poor productivity to consider possible worst case scenarios. However, we omitted this option to keep the focus on reservoir scale uncertainties.

Design parameters define those properties of the operation over which we have control – properties we can engineer or adapt. In contrast to uncertain parameters, which cannot be optimized, engineering parameters allow us to tune reservoir performance to our needs. Wells have many adjustable parameters including the location, the completion length and operational constraints such as flow rate or BHP. In this study we limit our focus on the borehole positions and how they are operated. Both wells in our doublet can vary within a wide range of locations, as illustrated in Fig. 4. Due to the full factorial design of experiments, the sampling of well locations is restricted to 6 locations per well to avoid an inflation of the computational effort. The re-injection temperature of the water may vary between 70 and 100 °C to reflect different modes of power plant operation. Table 2 summarizes the variable model parameters' range and their values sampled for proxy and response surface training.

3.4. Training simulations

When uncertainty in the geological model is considered, we may find

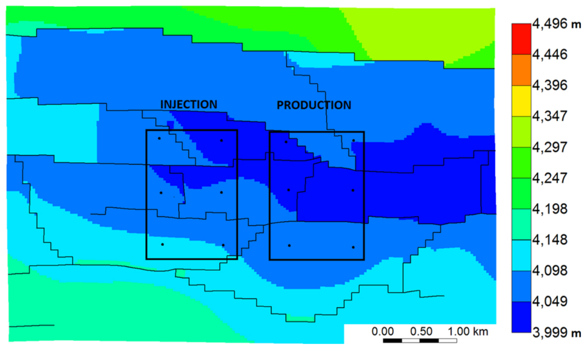


Fig. 4. Top surface of the geothermal Watt Field (depth in meters below ground level) and the range for borehole positions and locations sampled for proxy training within it.

Table 2
Variable parameters, their sampled values for proxy model training and their range.

Parameter	Sampled values	Range
Top surface model (TS)	1, 2, 3	1, 2, 3
Facies model (CO)	1, 2, 3	1, 2, 3
Fault model (FM)	1, 2	1, 2
Fault transmissibility	1%, 90%	0–100%
Re-injection temperature	70 °C, 100 °C	70–100 °C
Injector *X*-coordinate	21, 30	21 - 30
Injector *Y*-coordinate	14, 21, 28	14 - 28
Producer *X*-coordinate	38, 47	38 - 47
Producer *Y*-coordinate	14, 21, 28	14 - 28

Table 3
Geological scenarios and the respective number of model parameters sampled in a full factorial design.

Geological scenario	Fault transmissibilities	Re-injection temperatures	Injector locations	Producer locations	Number of simulations
CO1_FM1_TS1	2	2	6	6	144
CO1_FM1_TS2	2	2	6	6	144
CO1_FM1_TS3	2	2	6	6	144
CO1_FM2_TS1	2	2	6	6	144
CO1_FM2_TS2	2	2	6	6	144
CO1_FM2_TS3	2	2	6	6	144
CO2_FM1_TS1	2	2	6	6	144
CO2_FM1_TS2	2	2	6	6	144
CO2_FM1_TS3	2	2	6	6	144
CO2_FM2_TS1	2	2	6	6	144
CO2_FM2_TS2	2	2	6	6	144
CO2_FM2_TS3	2	2	6	6	144
CO3_FM1_TS1	2	2	6	6	144
CO3_FM1_TS2	2	2	6	6	144
CO3_FM1_TS3	2	2	6	6	144
CO3_FM2_TS1	2	2	6	6	144
CO3_FM2_TS2	2	2	6	6	144
CO3_FM2_TS3	2	2	6	6	144
Total					**2592**

well locations, which provide maximum heat extraction in one case, but fail to meet economic limits of heat production in another. Thus, reservoir performance needs to be assessed for every possible geological scenario. The total number of simulations from a full factorial design equals the product of the number of sampled points of all variables. In our study this equates to 3 top surface models × 3 facies models × 2 fault models × 2 fault transmissibilities × 2 re-injection temperatures × 2 injector *X*-coordinates × 3 injector *Y*-coordinates × 2 producer *X*-coordinates × 3 producer *Y*-coordinates; this results in 2592 simulation runs in total, or 144 simulations for each of the 18 geological scenarios (Table 3).

To describe the reservoir performance for each of the tested development plans the following simulation outputs are recorded for both the injection and the production well:

- **Water rate** to monitor the general productivity of the wells in [L/s].
- **Bottom hole pressure** to check if and when the pressure constraint is triggered in [kPa].
- **Energy rate** of the fluid stream based on the water rate and the enthalpy in [MW].
- **Cumulative energy** of the fluid stream based on the water rate and the enthalpy in [TWh].
- **Outlet water temperature** of the fluid stream based on the water rate and enthalpy in [°C].

The numerical reservoir simulations are performed using STARS (CMG, 2014), a commercial thermal reservoir simulator developed for the oil and gas industry with an equation of state for water properties. It uses an upwind weighted finite difference scheme with fully implicit time-stepping to solve the subsurface mass transport, as well as conductive and convective heat transport.

3.5. Optimization experiments

Based on the variable model inputs and the recorded outputs, a linear regression proxy model is generated for each training simulation (as described in Section 2) representing the response of each model as 6 parametric values (5 for the time series and 1 for pressure). Next, a GPR response surface model is trained to predict the linear regression proxy model exponents (6 outputs) for each of the 18 geological scenarios listed in Table 3 to allow the prediction of the time series data. The GPR response surfaces are then used for the following optimization experiments, which consider the impact of MOO for a geothermal reservoir for a range of objectives:

Exp 1: MOO–Min(dBHP) vs. Max(net cumulative enthalpy)
MOPSO optimizes the response surface for a single geological model (***CO1 FM1 TS1***), by evaluating the trade-off between maximizing the net cumulative energy extraction over 60 years (cumulative enthalpy at outlet–inlet), and minimizing the difference in BHP (dBHP) between the injection and production well. The former objective accounts for the efficiency of heat extraction at the surface or the sustainable amount of available energy. The latter reflects the energy demands in re-injection (pump costs) where an increased draw-down in the production well compels a comparable increase in re-injection rate achieved through increased pumping.

Exp 2: MOO–Min(dBHP) vs. Max(net cumulative enthalpy)
As above, but for geological scenario ***CO2 FM2 TS2***.

Exp 3: MOO–Min(dBHP) vs. Max(net cumulative enthalpy)
As above, but for geological scenario ***CO3 FM2 TS3***.

Exp 4: MOO–Min(Mean dBHP) vs. Max(Mean cumulative enthalpy)
As above, but the optimizer is steered by the mean response of all geological scenarios. Each iteration from MOPSO evaluates across all 18 geological proxy models, providing a mean dBHP and net enthalpy value.

As we have created a total of 18 GPR models (one per geological scenario), our total number of 2592 simulation responses represents only 144 simulation runs per geological scenario. The 18 GPR models require validation to make a judgment on their usefulness in prediction. As such, 80% of the 144 simulations are used to train the GPR model and 20% are held back for validation, a commonly used split in training/validation.

For each experiment the MOPSO algorithm iterates 500 times. The results of the proxy building and validation along with the results of optimization are provided below.

4. Results

4.1. Proxy testing and validation

Fig. 5 shows the predicted vs. true responses for one of the geological models, *CO3 FM2 TS3*, where the red line indicates perfect predictability of the model response by the regression model and the blue dots show the actual discrepancy in prediction. The results (blue dots) show a good level of fit and predictability for most data, but with some significant discrepancies between the proxy and true values for the test data, particularly for coefficient 2 and 3 (cf. Fig. 5 top-middle and top-right), which are used to fit the production temperature time series data. This means that in some regions of the parameter space the response surface has difficulties to correctly predict the respective proxy model for the time series of production temperature and that some errors have to be expected.

The errors in prediction can result from distinct changes of heterogeneously distributed parameters like permeability and porosity. Better predictive accuracy could be achieved through denser sampling of the parameter space in these specific regions. In general, more dense sampling also significantly increases the computational effort. This highlights the importance of the experimental design, which determines the way the parameter space is sampled. In this case, however, it is difficult to predict the regions that need denser sampling *a priori*, as the original Watt Field porosity and permeability models where populated using stochastic methods.

The complex nature of the response surface is further illustrated in Fig. 6, which shows the variation in the 6 linear regression model parameters (model input) with respect to the associated cumulative enthalpy production (model response) for all 144 simulation runs of geological scenario *CO3 FM2 TS3*. These variations mirror observations from, e.g. (Crooijmans et al., 2016; Willems et al., 2017b; Nick et al., 2016) that borehole placement and associated enthalpy production is affected by heterogeneity, which regulates reservoir productivity (and forecasting) in geothermal systems. Validation runs, again, are mostly well predicted, with some notable exceptions in the coefficients 2 and 3. The influence of heterogeneity (porosity and permeability) and faults (that may or may not be sealed) compounds to create a complex model response that is hard to capture as a response surface.

To visualize the impact of discrepancies in the response surface shown in Fig. 6, the worst fitting models (models with a maximum error in any time step of over 1.5 °C) are examined in Fig. 7 for the same geological scenario *CO3 FM2 TS3*. As the response surface predictions sometimes undercut or overshoot the validation data, the error changes over time. The observable trend is most likely related to the regression model used for the proxy training. The shape of the training data is not perfectly described by a linear plateau and a subsequent quadratic function.

Fig. 7(a) represents the error over time between predictor and simulation, while Fig. 7(b) shows the temperature predictions from both the response surface approach (solid lines) and "true" simulation outputs (dashed lines). Clearly, in some cases the errors are more considerable with a deviation of up to ∼6 °C deviation, which represents a noticeable forecast error. The total number of poorly predicting models accounts for 8 of 144 development scenarios or around 5% of predictions. Despite these errors, the response surfaces still correctly predict the general trend and are still sufficiently accurate to be used in optimization experiments.

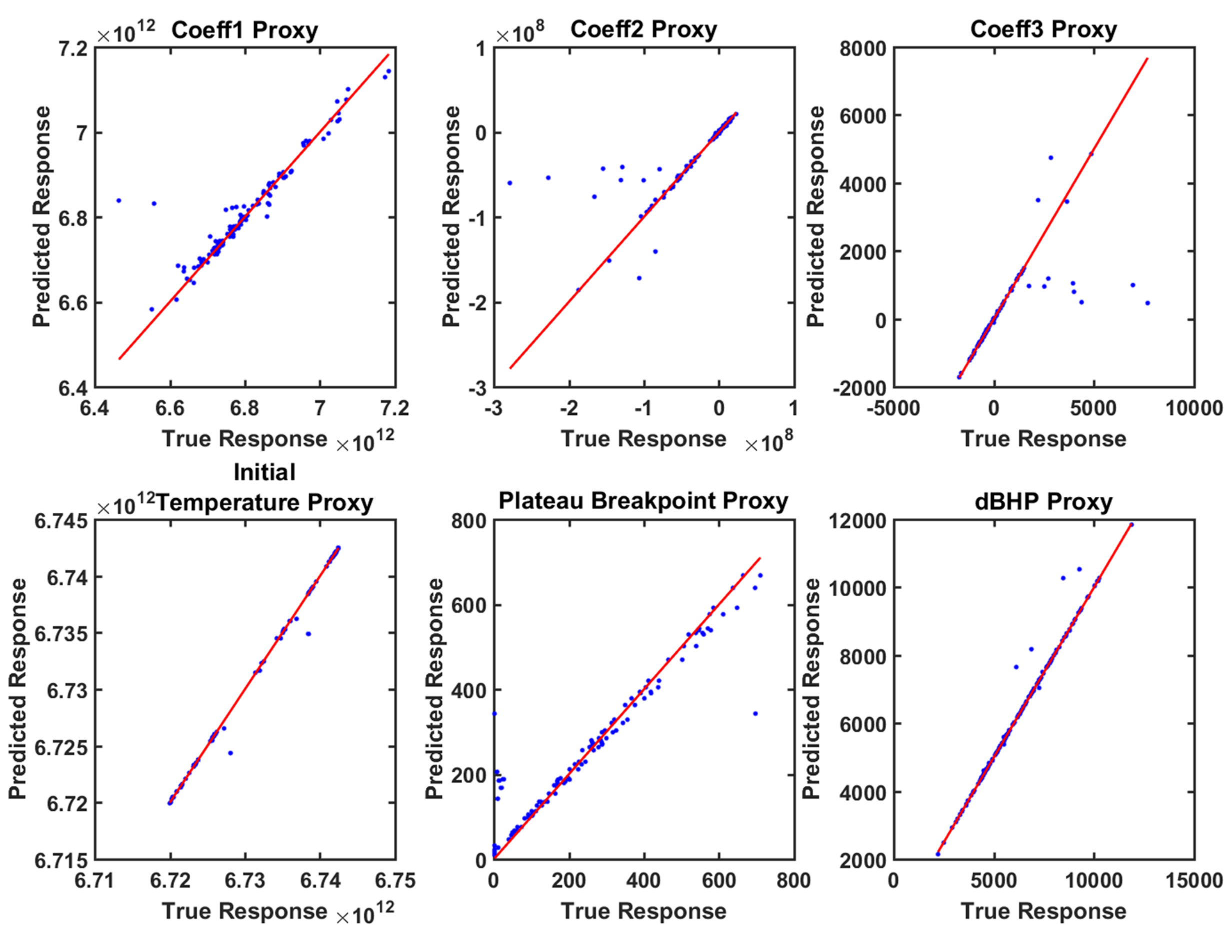


Fig. 5. Proxy validation results for geological scenario CO3 FM2 TS3. Cross plot of true vs. predicted response for the 6 proxy model coefficients to predict outlet temperature and pressure. (For interpretation of the references to color in this figure citation, the reader is referred to the web version of this article.)

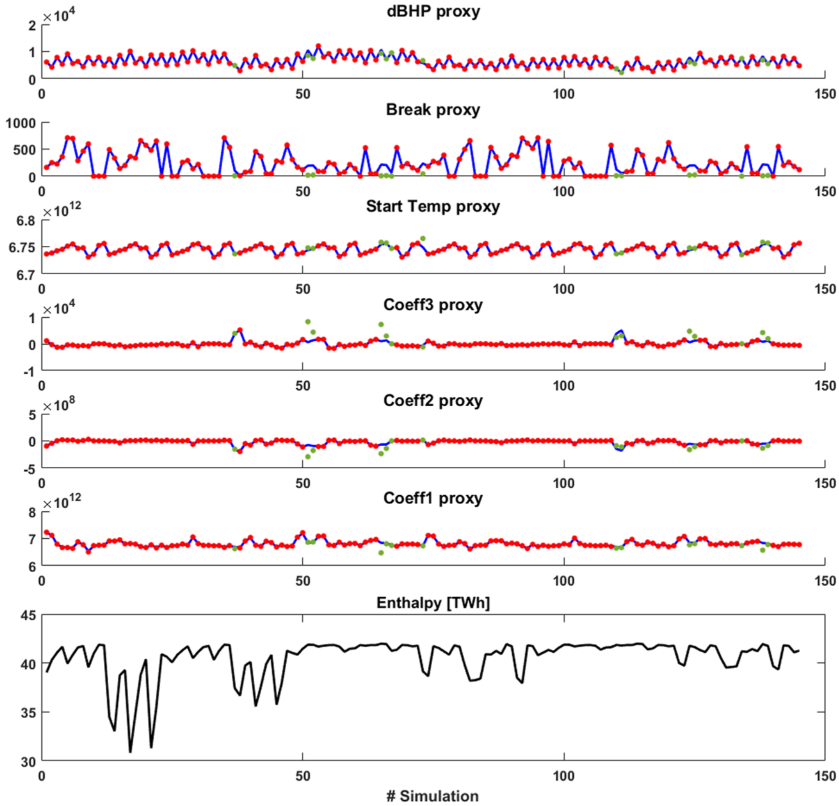


Fig. 6. Validation results for geological model *CO3 FM2 TS3* showing the fit of all 6 response surfaces used in predicting outlet temperature time series (as per Fig. 1) and the dBHP with their associated cumuluative enthalpy production. The blue curves represent the Gaussian process regression model fit to simulation training data (red dots). The validation data set is shown as green dots and demonstrates a range of quality of predictability. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Optimization experiments

Experiments 1–4 results' are shown in Fig. 8 plotting net cumulative enthalpy vs. dBHP. The full 2000 iterations (500 × 4 experiments) show as gray dots and the respective Pareto fronts show as colored dots, where different colors represent the four experiments. The shape of the Pareto front evolves from the simulation responses and shows the trade-off between increasing enthalpy and the increased dBHP or energy requirement to achieve the increased enthalpy. We see in all cases that there is a non-linear relationship between increasing enthalpy with increasing dBHP such that initially small increases in dBHP yield large increases in enthalpy, but this trend stops abruptly such that the highest enthalpy values require large dBHP values and therefore large pumping requirements.

The Pareto fronts show the impact of the geological model, where the dBHP value varies greatly between the different geological scenarios. For optimization under uncertainty (blue dots) the Pareto front is located between the univariate runs and demonstrates the influence of averaging the results across many scenarios.

Fig. 9(a) shows the locations of the wells for both the injectors and producers of the Pareto front models from optimization under uncertainty. Locations are depicted as circles with their respective re-injection temperature, which varies between 70 and 100 °C. The corresponding production response is redrawn in Fig. 9(b). Three doublets are highlighted as red squares, blue triangles and green triangles. In comparison, these three development options yield a similar response in terms of cumulative enthalpy and dBHP, whereas their well configurations are significantly different, as they cover a range of locations.

To test the proxy we examined the predictability of the response surface against the Pareto front models from MOO on three geological scenarios *CO1 FM1 TS1*, *CO2 FM2 TS2* and *CO3 FM2 TS3*. Fig. 10 shows the predicted (solid line) vs. actual (dashed line) outlet temperature forecasts for a number of development options across these three geological scenarios. The pink, yellow and blue lines show randomly

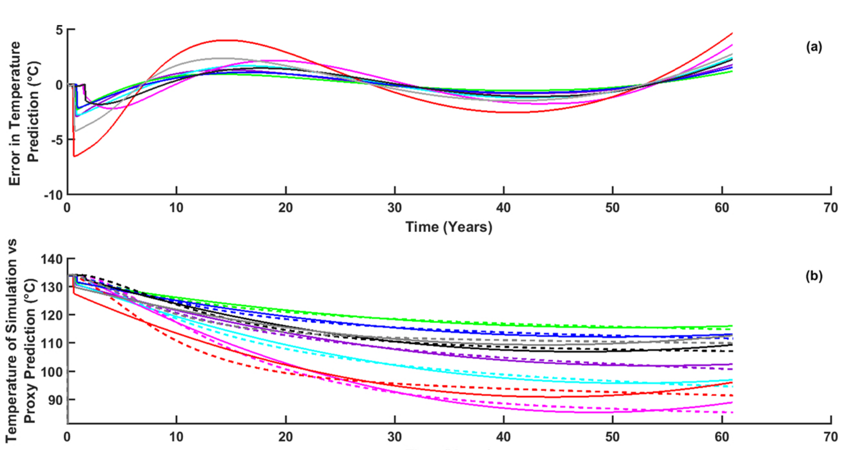


Fig. 7. Response surface prediction error for the worst fitting proxy models for the validation data set. (a) shows the error over time in temperature prediction for the 8 worst models with maximum errors above 1.5 °C. (b) shows the outlet temperature of the response surface (solid lines) vs. simulation result (dashed lines) for these cases.

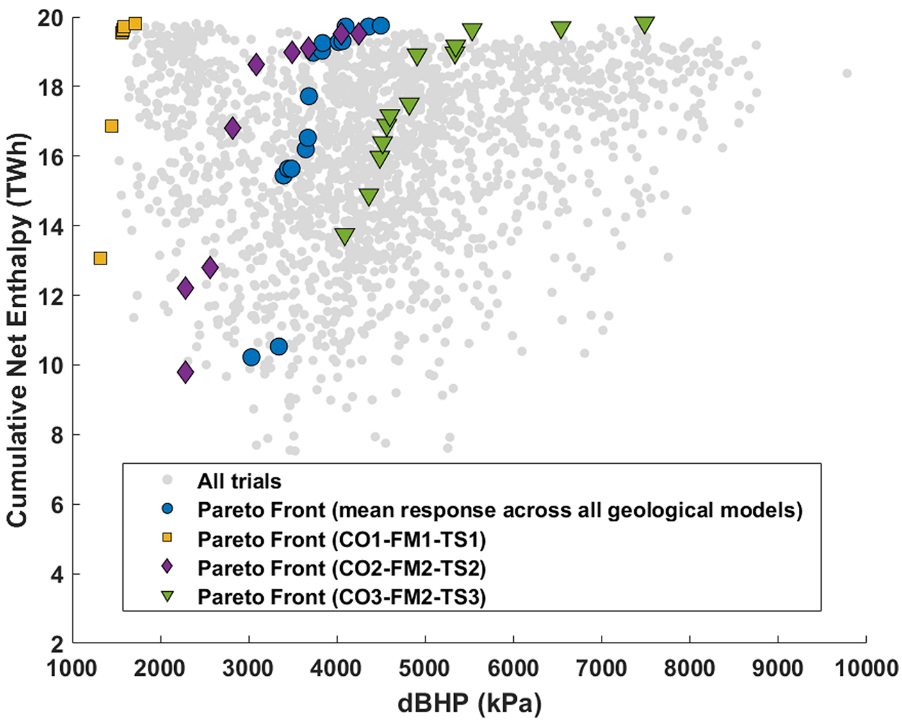


Fig. 8. MOO results for 3 geological scenarios and the optimization under uncertainty with their respective Pareto fronts (yellow: *CO1 FM1 TS1*, purple: *CO2 FM2 TS2*, green: *CO3 FM2 TS3*, blue: *optimization under uncertainty*). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

selected development options for the three geological models, which all show a good fit between the actual and predicted temperature profiles and generally represent the quality of fit for most cases. The green lines show the same scenario as the blue lines but for a lower fault transmissibility (green = 0.01 and blue = 0.9) supporting that predictability is also good with respect to fault transmissibility. The red line shows the worst fitting model from the validation set, taken from geological scenario *CO3 FM2 TS3*. The response surface for this geological scenario performs generally worse than the others tested, probably due to the impact of the permeability distribution and more complex fault model in this scenario. Over all models on the Pareto front used to test the response surface predictability, the forecasts generally show a good fit to the actual predictions of thermal output at the production well. However, some significant discrepancies occur due to the difficulty of fitting the subsets of 144 training simulations to their respective response surfaces when the corresponding geological scenarios have very different levels of geological complexity as illustrated by the red line. It shows the largest observed discrepancy from the Pareto front of over 15 °C by the end of the forecast time period, demonstrating that issues exist in fitting a response surface to some geological scenarios more than others. The inconsistency in fit quality between geological scenarios will impact upon the mean response of the ensemble where proxy errors are significant and/or where multiple errors occur together to skew the mean/variance. As geological features like faults will have more of an impact in some geological scenarios, the sufficiency of sampling to generate a robust response surface will diminish and with it the predictability (as shown in Zubarev, 2009). Thus, such scenarios require denser sampling for

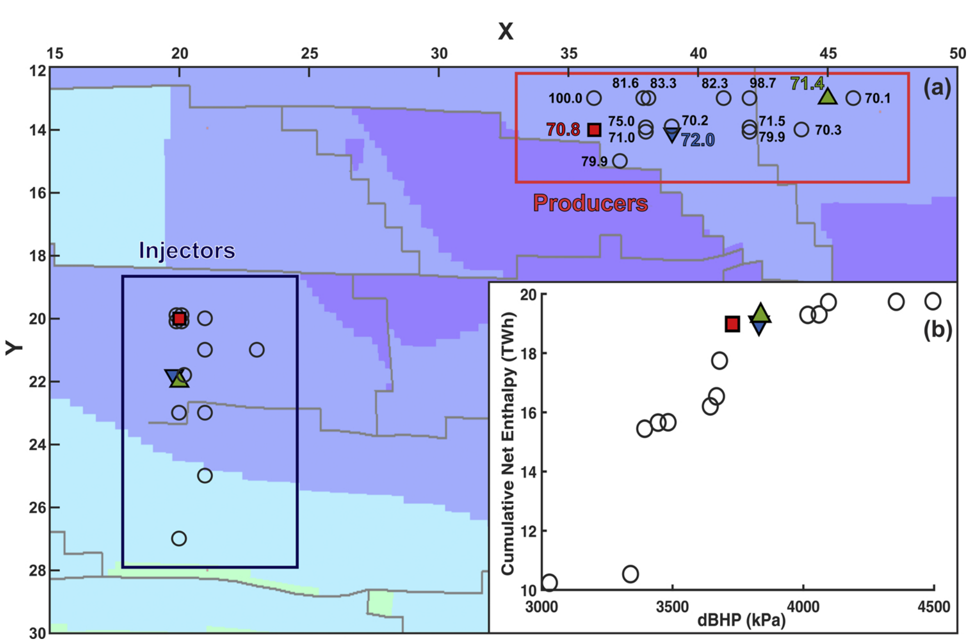


Fig. 9. Top surface of the geothermal Watt Field (color-coding for depth in reference to Fig. 4) and the *X*/*Y* well locations of the optimization under uncertainty Pareto front model (a) and corresponding Pareto front (b); the three highlighted doublets with significantly different well locations exhibit very similar responses. Numerical values associated with the producers represent the reinjection temperature chosen by the optimizer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

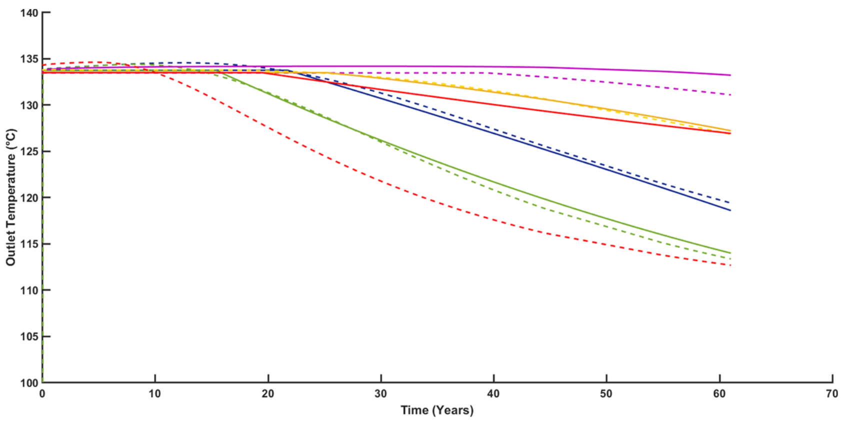


Fig. 10. Predicted (solid line) vs. actual (dashed line) outlet temperature for a set of randomly selected models on the Pareto front across geological scenarios *CO1 FM1 TS1*, *CO2 FM2 TS2* and *CO3 FM2 TS3*. Pink, yellow and blue lines represent a model from each of the geological scenarios. Green shows the same setup as blue, but using a fault transmissibility value of 0.01 (instead of 0.9). The red line shows the worst fitting model from the Pareto front of *CO3 FM2 TS3*. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

training the response surface appropriately and reduce the impact of outliers on optimization performance.

As a comparison with the MOO results, we ran a single objective optimization (SOO) using PSO for maximizing cumulative net enthalpy, and minimizing dBHP separately to evaluate the outcome when only a single objective and a single geological model was considered. PSO was carried out for 1000 iterations for each geological scenario *CO1 FM1 TS1*, *CO2 FM2 TS2* and *CO3 FM2 TS3* resulting in 6 runs of PSO (3× for cumulative net enthalpy and 3× for dBHP) producing a total of 6000 iterations across the response surface models for the appropriate geological scenario. The results of the single objective runs are given in Fig. 11 which shows the optimal well placements for both injectors (I) and producers (P) for each geological model and for each objective (1–3 = maximizing net enthalpy, 4–6 = minimizing dBHP). Maximizing enthalpy drives the wells to maximize their distance to offset thermal breakthrough. Thus, all geological models show wells widely separated with similar inter-well distances of approximately 2.5 km. The table as part of Fig. 11 lists the objective scores for the optimized scenario shaded in blue. The values of the same objectives for the same well locations/ setups for the other two geological scenarios are also provided along with the coefficient of variation (CV, i.e. ratio of the standard deviation to the mean) for the development scenario. The results show generally small CV's for net enthalpy, even for most cases where the dBHP was optimized (wells 4–6).

dBHP shows a preference for a closer well spacing but the value of dBHP and the optimal well location are more variable, dependent on the geological scenario selected; for dBHP the optimization is much more sensitive to the geological model. The overall net enthalpy for SOO that minimizes the dBHP is significantly lower than for any of the options discovered in MOO. Hence, optimizing only on this objective would result in a much reduced thermal output of the system, i.e. a lower enthalpy rate and a reduced cumulative enthalpy over time.

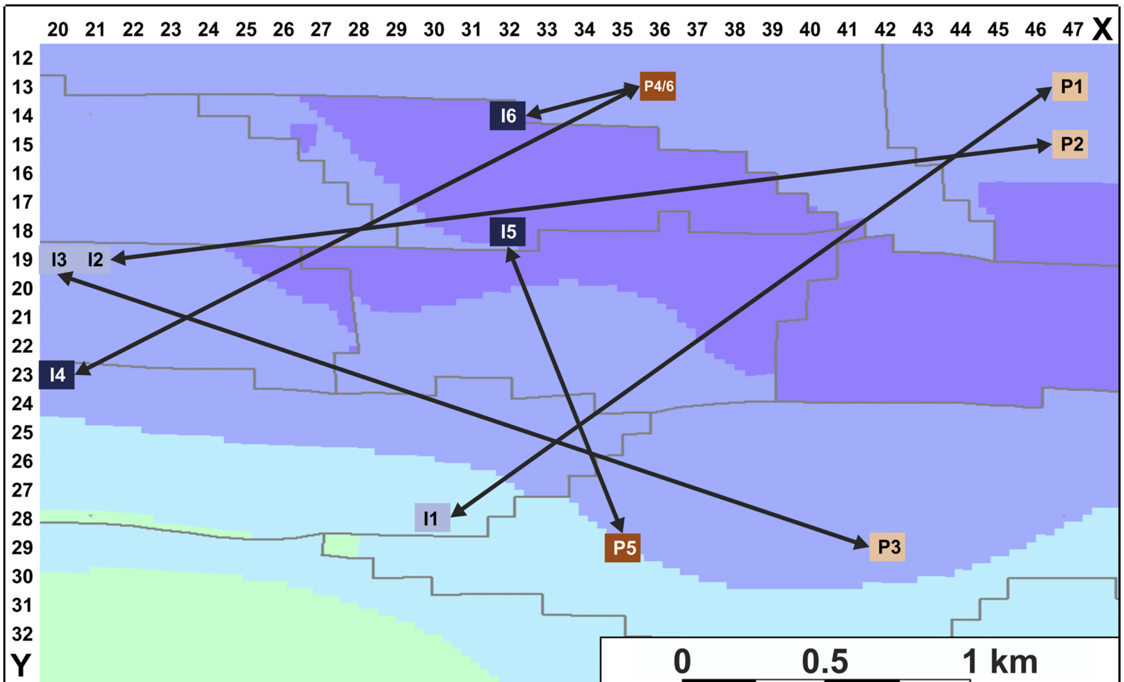
5. Discussion

This work shows that combining MOO and optimization under uncertainty methods on a realistic low enthalpy geothermal reservoir provides a way to explore the engineering trade-offs that control the profitability of a geothermal development and do this while accounting for risk. Optimization shows that a range of development scenarios exist along the trade-off surface that are quite variable in terms of well location/inter-well distance (1600–3000 m) and re-injection temperature (70–100 °C). Thus, including uncertainty into MOO by optimizing on the mean response of many models requires additional steps and simulation effort in prediction (in this case to generate the proxy models and response surfaces). However, the big variations in dBHP between geological scenarios (> 3000 kPa in dBHP as shown in Fig. 8) demonstrate the need to account for uncertainties in optimization of heterogeneous geothermal systems.

The key comparison between single and multi-objective outcomes shows the importance of considering both objectives simultaneously, particularly for the dBHP case where net enthalpy is significantly reduced. The results of SOO on maximizing enthalpy output show a preference to maximize well spacing, which obviously increases the time to cold water breakthrough. What we have not considered in this work, and would form an interesting additional study, would be to look at objectives that include the constraints from well spacings, namely the additional costs of extending the length of wells, additional pumping costs for larger inter-well distances and the reduction in the density of wells that may be used to extract energy (reduction in the overall resource energy rate of recovery). Well costs for varying inter-well distances has been looked at by for instance Willems et al. (2017a) where NPV is improved by reducing the well spacing as well costs drop from shorter drilling distances.

Overall, the predictive performance of our approach of running optimization on response surfaces is good. The models of the geological scenarios show reasonable predictability and the GPR approach shows significant computational gains over direct simulation. However, some validation points for some geological scenarios on the Pareto front show proxy performance is potentially insufficient to capture the true impact on reservoir performance for different well options given the reservoir heterogeneity. In addition, increasing the number of geological scenarios for a defined number of simulation runs may have an effect on the ability to fit a regression model as the number of training points per proxy reduces or as different geological scenarios must be represented by more complex response surfaces. For instance, the scenarios developed herein with a high discrepancies between the response surface predictions and full-physics simulation, could be explained by the presence of more complex structural models which impact on the quality of model fit. This may explain why similar work in CCS by Oladyshkin et al. (2011) has shown good results using a similar polynomial proxy, but where the fluid (CO_2) mobility is significantly more favorable. Thus, the so called “flora's rule” applies (Ringrose and Bentley, 2015) reducing the impact of heterogeneity and the geological model used (Class et al., 2009) is quite simple in comparison with the Watt field. As observed in other industrial applications, proxy models may produce erroneous forecasts and for the case of geothermal where margins for economic success are quite narrow, care must be taken to balance potential time savings vs. forecast robustness.

The computational performance is mostly controlled by the number of simulation runs in the design of experiments stage to seed the response surface models, which run in seconds. The initial 2592 reservoir simulations took around 8000 CPU hours to run (3+ hours per simulation) which were spread across multiple cores and CPUs in a small compute cluster, reducing the time to a few days. The multi-objective runs that estimated the averaged response from 18 geological models would have taken 54 h per iteration, or 27,000 CPU hours per MOO run. Thus once built, the response surface approach



		Cumulative Net Enthalpy (TWh)			CV	dBHP (kPa)			CV
	distance (m)	111	222	323		111	222	323	
I1 - P1	2202	**3.99**	3.98	3.99	0.06%	**786**	846	1187	23.00%
I2 - P2	2631	3.97	**4.00**	3.99	0.49%	700	**565**	936	25.61%
I3 - P3	2416	3.80	3.88	**4.00**	2.64%	543	451	**1003**	44.40%
I4 - P4	1887	**1.96**	2.05	2.06	2.83%	**135**	555	641	60.96%
I5 - P5	1170	1.87	**1.89**	1.77	3.59%	776	**141**	1146	73.96%
I6 - P6	412	1.78	1.83	**0.40**	60.96%	273	533	**327**	36.36%

Fig. 11. Top surface of the geothermal Watt Field (color-coding for depth in reference to Fig. 4) and the well locations of optimal solutions from SOO to maximize enthalpy and minimize dBHP on geological scenarios *CO1 FM1 TS1*, *CO2 FM2 TS2* and *CO3 FM2 TS3*. I1-P1 to I3-I3 show the locations for the injectors (I) and producers (P) of the maximization (enthalpy) runs while I4-P4 to I6-P6 show the well locations for the minimization (dBHP) runs. The table at the bottom compares the results of the optimization process for both enthalpy and dBHP (shaded bold-print values) to the respective values for the other geological scenarios. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

has many performance advantages. Moreover, the response surface models could be integrated into simulations with a wider scope investigating plant operation or economics as a sub-model routine. Further work to reduce the computational load would be to reduce the number of geological models from 18, thereby allowing fewer reservoir simulation runs overall or allowing more training runs for the proxy model (up from 144) to improve predictability.

6. Conclusion

In summary:

1. Geological heterogeneity has a large impact on geothermal reservoir productivity and different geological scenarios would have different optimal development plans.
2. Multi-objective optimization can help uncover the trade-offs in any competing reservoir objectives such as maximizing heat production while minimizing the energy requirement in pumping.
3. Optimization under uncertainty helps identify the best development option on average given an ensemble of geological scenarios.
4. Response surface models (built using methods like GPR) can improve the efficiency of forecasting such that large numbers of geological scenarios can be considered.
5. Once trained, response surface models could augment economic models or system models that focus on plant operation.
6. The presented method is generally applicable to other reservoir types, as well.

Authors' contribution

Daniel Schulte: conceptualization, methodology, software, formal analysis, visualization, writing – original draft, writing – review and editing. Dan Arnold: conceptualization, methodology, software, formal analysis, writing – original draft, writing – review and editing. Sebastian Geiger: supervision, resources, writing – review and editing. Vasily Demyanov: methodology. Ingo Sass: resources, funding acquisition.

Conflict of interest

None declared.

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